

# Lorenzo Cardoni

📍 Ancona (Available for transfer)    ✉ cardonilorenzo31@gmail.com    ☎ +39 3402655536    in Lorenzo Cardoni    🔗 Lorenzo-Cardoni

## Technical Skills

- **Programming Languages:** Python, C/C++, MATLAB, SQL
- **Libraries and Frameworks:** Pandas, NumPy, scikit-learn, SciPy, Matplotlib, YOLO, TensorFlow, OpenCV, Simulink
- **Tools & Other Technologies:** Arduino, Raspberry, STM32 Devices, Git, Latex

## Education

**Master's Degree in Computer and Automation Engineering** - Università Politecnica Delle Marche Oct. 2022 – Feb. 2025

- **Degree Grade:** 110/110 cum Laude
- **Thesis Title:** [Hardware-In-The-Loop approach for fault simulation and validation of diagnostic modules in UAV](#) 🔗
- **Thesis Advisor:** Prof. Alessandro Freddi

**Bachelor's Degree in Computer and Automation Engineering** - Università Politecnica Delle Marche Sept. 2019 – Dec. 2022

- **Degree Grade:** 103/110
- **Thesis Title:** [Study and Development of linear controllers for Attitude Control of Mini-Drones](#) 🔗
- **Thesis Advisor:** Prof. Gianluca Ippoliti

## Experience

**Junior Data Engineer**, Lega del Filo d'Oro Ancona, Italy  
April 2025 – Oct. 2025

- Designing robust ETL pipelines using *Python* and *SQL*, ensuring accurate and efficient data flow to support business intelligence and reporting workflows.
- Integrating an *LLM* with various business data sources to dynamically generate BI visualizations.

**Laboratory Technician**, Università Politecnica Delle Marche Ancona, Italy  
Jan. 2024 – March 2024

- Design and development of an embedded prototype for detecting and estimating the position of underwater objects, validated through real-world testing.
- I led the project coordination with the client using the *AGILE methodology* and successfully achieved *Technology Readiness Level 5* (TRL 5).

## Scientific Publications

**Benchmark Analysis of YOLOv8 for Edge AI Video-Surveillance Applications** Oct. 2024

Daniele Berardini; Lucia Migliorelli; **Lorenzo Cardoni**; Christian Parente; Alessandro Rongoni; Daniele Sergiacomi; Adriano Mancini

**DOI:** [10.1109/MESA61532.2024.10704889](https://doi.org/10.1109/MESA61532.2024.10704889) 🔗

## University Projects

**Development of a toolbox for Fault Injection and Implementation of a Machine Learning Model for Fault Detection in a PX4 HTL for an Unmanned Aerial Vehicles (UAV)** Nov. 2024 – Jan. 2025

- Developing a *machine learning*-based fault diagnosis system for UAVs in *MATLAB* & *Simulink*, validated through *Hardware-in-the-Loop (HITL)* simulations on the Flight Controller.
- Integrating diagnostic modules into simulation environments and real-time embedded platforms for performance assessment.

**Implementing a Convolutional Neural Network (CNN) on an STM32 board** March 2024 – May 2024

- Training a *CNN* for *image recognition* using *TensorFlow*, implementing it on an *STM32F429I-DISCO* board for real-time inference.
- Validating the CNN's performance on the STM32 board.

**Study of feature super-resolution techniques for Edge AI Video-Surveillance Applications** July 2023 – Sept. 2023

- Conducting a benchmark study of super-resolution techniques in *Python* to improve small object detection performance of *YOLOv5 model* on edge devices.
- Developing a flexible framework enabling integration of generative models for feature enhancement during *object detection* inference.

## Challenges

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### Hackathon UNIVPM 2023 & 2024

Mar. 2023 & Mar. 2024

- 2023: Developing a website for managing large tourist events.
- 2024: Training a deep learning model for recognizing waste using camera input.

### Bosch Future Mobility Challenge

Nov. 2022 - March 2023

- Participating in the Bosch Future Mobility Challenge, developing AI algorithms for the autonomous driving of a 1:10 scale vehicle, focusing on *object detection* with a *YOLOv5 model*.

## Languages

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- Italian, Native Speaker
- English, B2 (Fluent)

I authorize the processing of personal data present in the CV according to the Regulation of the European Parliament 679/2016.

Ancona, January 19, 2026

**Lorenzo Cardoni**

